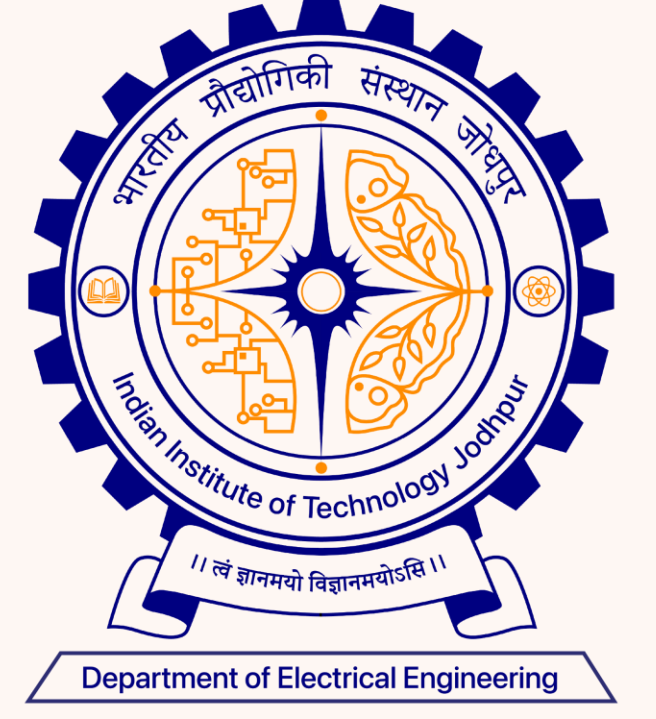


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A Study of Deadzone Structure for Multi-Dimensional Haptic Force Perception

Ravi Prakash Meghwanshi, Amit Bhardwaj*, Himanshu Kumar

*email: amitb@iitj.ac.in

Indian Institute of Technology Jodhpur, Rajasthan, India

Introduction

- A perceptual adaptive sampling approach, based on Weber's law of perception, has been used to reduce haptic data in TPTA systems.
- 1D perceptual deadband: $(1 - \delta)F_{n-1} < F_n < (1 + \delta)F_{n-1}$

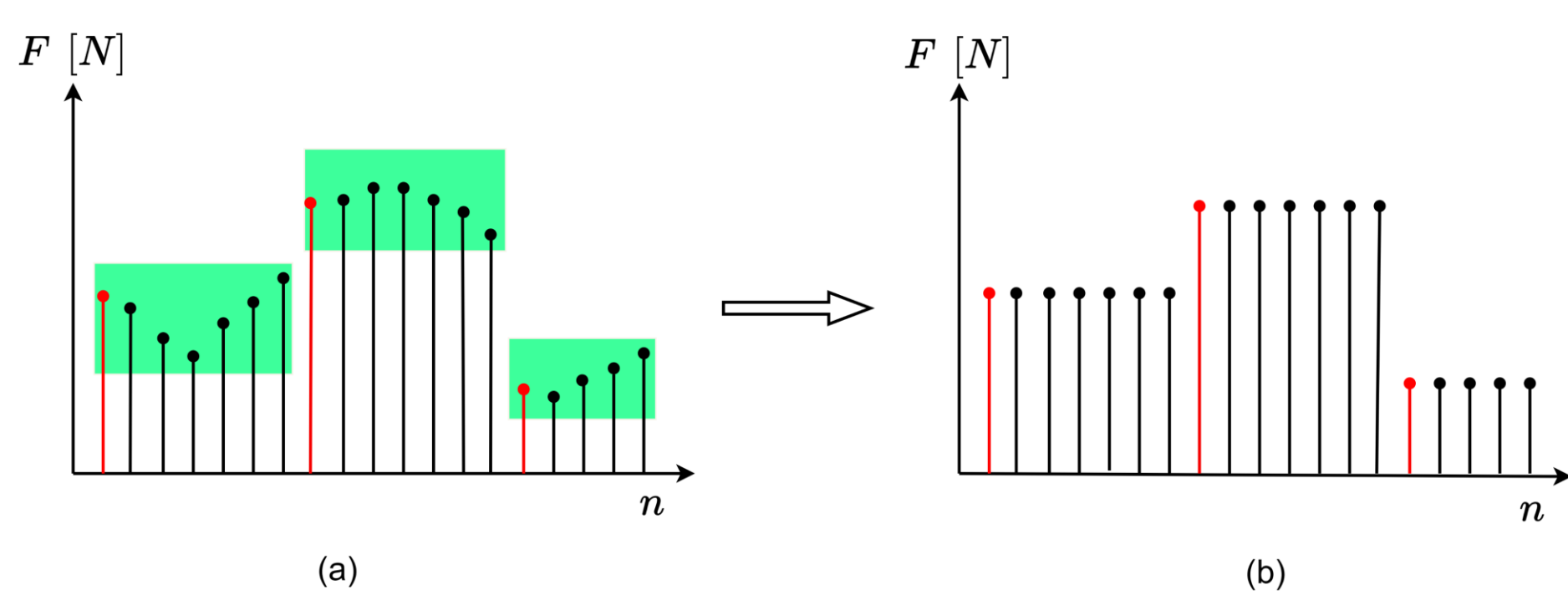


Fig. 1: An illustration of the 1D perceptual deadband approach at
(a) Transmitter side: green region denotes the perceptual deadband
(b) Receiver side: Signal reconstruction using the sample-and-hold technique.

Perceptual Deadzone

$$\text{Classification error: } E^k = \frac{1}{N} \sum_{i=1}^N \left(w_p \frac{(1 + y_i)}{2} + w_{np} \frac{(1 - y_i)}{2} \right) \mathbf{1}(y_i \neq h^k(r_i, d_i))$$

1. Generalized Deadzone: RBF kernel-based SVM

2. Elliptical Deadzone:

$$G(\vec{F}_t, \vec{F}_r, \theta) = \frac{((F_t^x - F_r^x) \cos(\alpha + \theta) + (F_t^y - F_r^y) \sin(\alpha + \theta))^2}{a^2} + \frac{((F_t^x - F_r^x) \sin(\alpha + \theta) - (F_t^y - F_r^y) \cos(\alpha + \theta))^2}{b^2} \geq 1$$

$$h_e(\vec{F}_t, \vec{F}_r, \theta) = \begin{cases} 1 & G(\vec{F}_t, \vec{F}_r, \theta) - 1 \geq 0 \\ -1 & G(\vec{F}_t, \vec{F}_r, \theta) - 1 < 0 \end{cases}$$

- Special case: Circular Deadzone ($a=b$)

$$C(\vec{F}_t, \vec{F}_r) = \frac{(F_t^x - F_r^x)^2 + (F_t^y - F_r^y)^2}{r^2} \geq 1$$

Methodology

- Machine learning classification-based approach
- Paradigm: Participant is exposed to a 2D kinesthetic force stimulus and is asked to react to the change.

Setup

- PHANTom Premium 1.5 HF haptic device
- Two-dimensional (2D) haptic force stimulus $\rightarrow$ +XY-plane

Stimuli

$$\vec{F}(t) = \begin{cases} \vec{F}_r \times 10t & \text{for } 0 < t \leq 0.1 \\ \vec{F}_r & \text{for } 0.1 < t \leq 2.1 \\ \vec{F}_t = \vec{F}_r + \Delta\vec{F} & \text{for } 2.1 < t \leq 4.1 \\ 0 & \text{for } t > 4.1 \end{cases}$$

$$\vec{F}_r = 3\angle 45^\circ, \Delta\vec{F} = [0, 1]N, \phi = [0^\circ, 360^\circ]$$

Data-Collection

- Participants: 12 users
- Total trials: 1000 trials per user

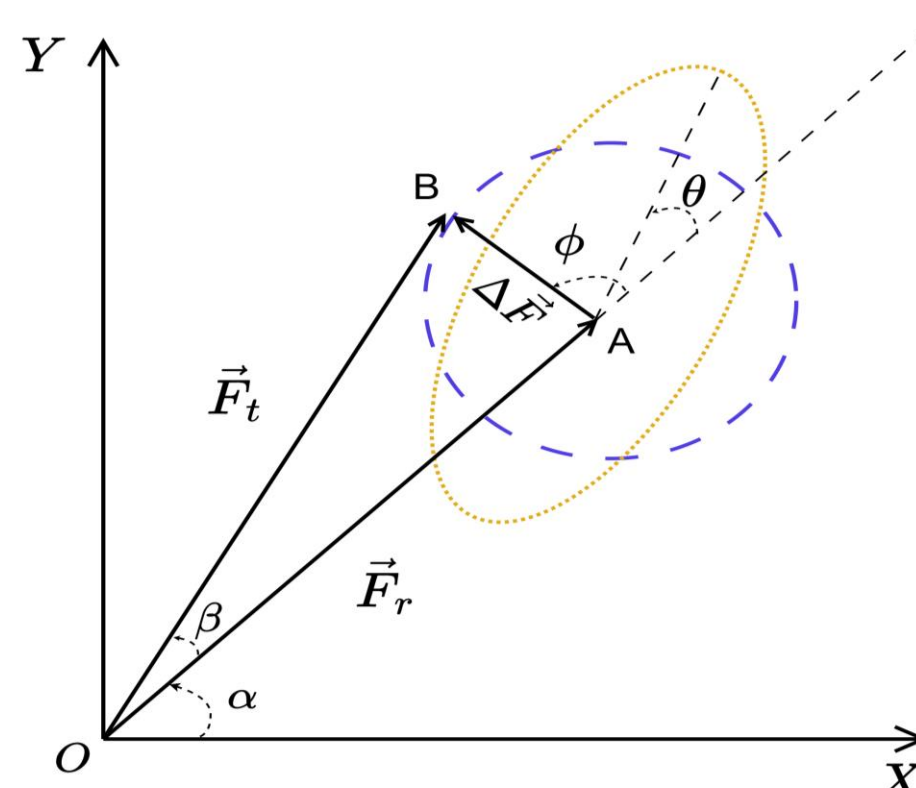


Fig.2: A possible 2D perceptual deadzone structure around the 2D reference force stimuli

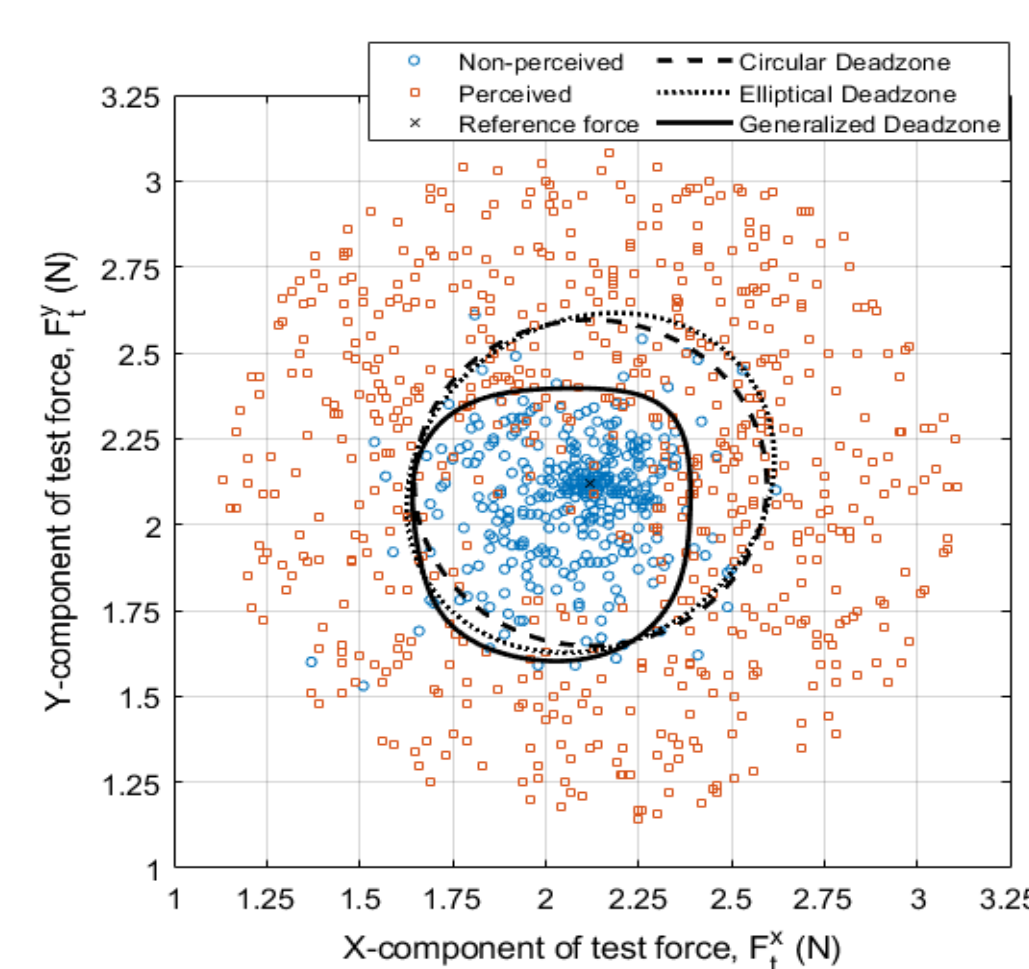


Fig. 3: Scatter plot for one participant

Results and Analysis

1. Shape:

- Area is statistically not different
- Avg. Errors: Ec, Ee, Eg are 28.98%, 21.03%, 16.68%
- Generalized deadzone is the optimal

2. Effect of Direction:

- Significant effect of force direction (ϕ) on JND, $F(15, 176) = 11.97, p < 0.001$

3. Symmetric Behavior:

- Asymmetric with respect to the reference force

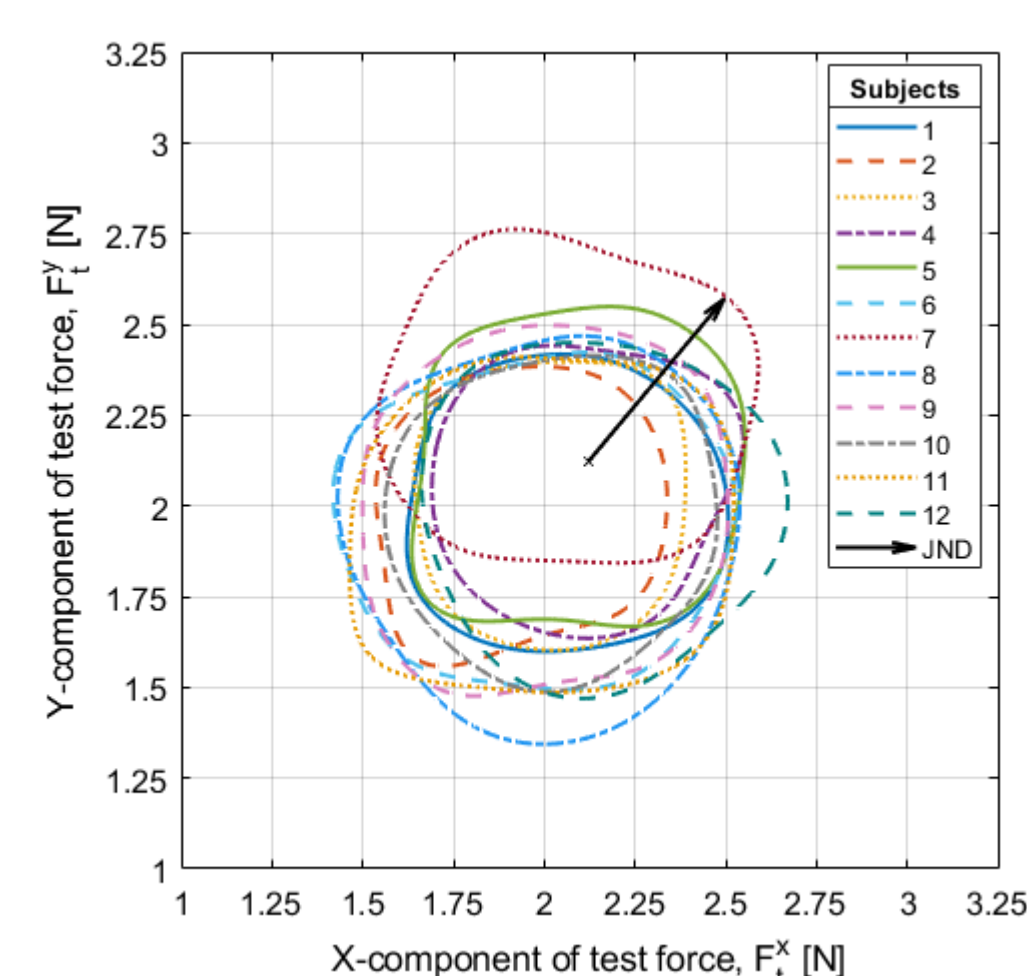


Fig.4: Optimal boundaries of the generalized deadzone

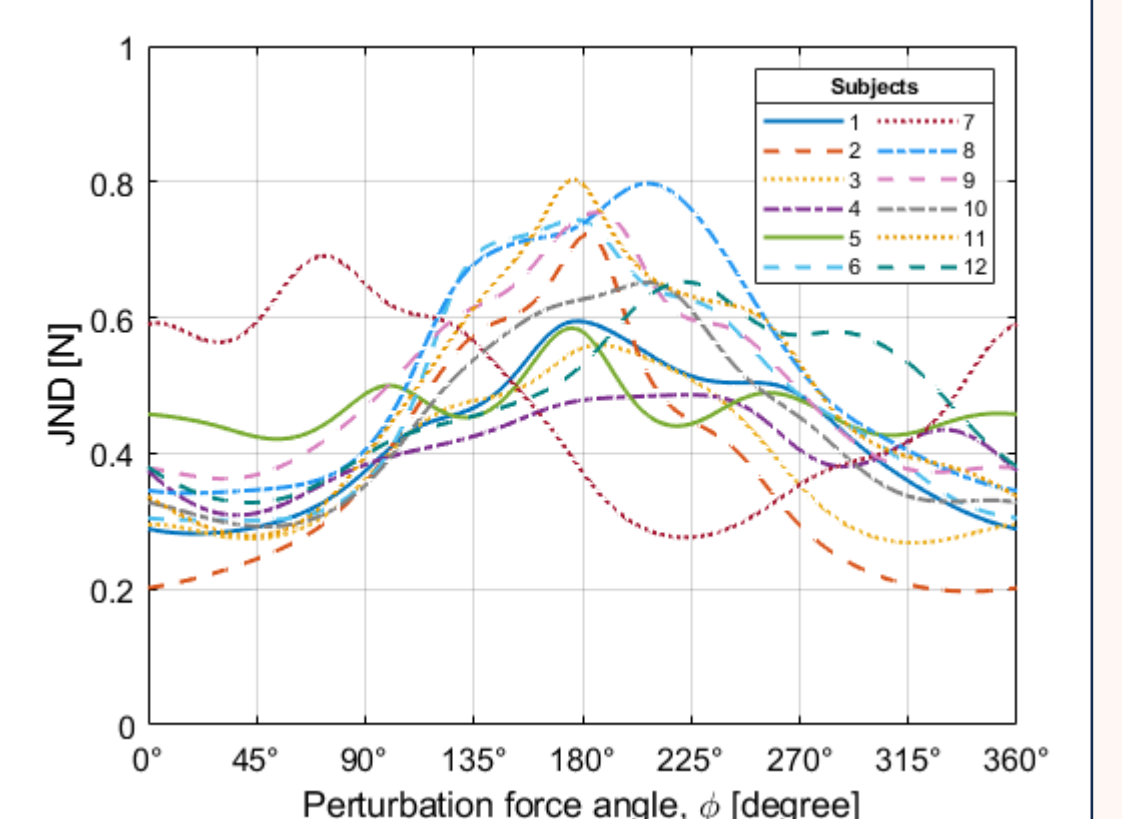


Fig.5: Force JND vs perturbation force angle (ϕ)

Discussion & Conclusion

- 2D Deadzone is **direction variant** and **radially asymmetric**
- Although approximately circular, the deadzone is **not centered** around the reference force.
- δ min**: in the reference force direction, **δ max**: along the perturbation direction ($\phi = 180^\circ$)
- In addition, the optimal deadzone structure also shows that humans are more sensitive to force perception if the change is made along the reference force direction.
- These findings may support future **perceptually adaptive haptic communication standards, coding/compression approaches, and interoperability frameworks** by enabling direction-aware force representation and transmission.

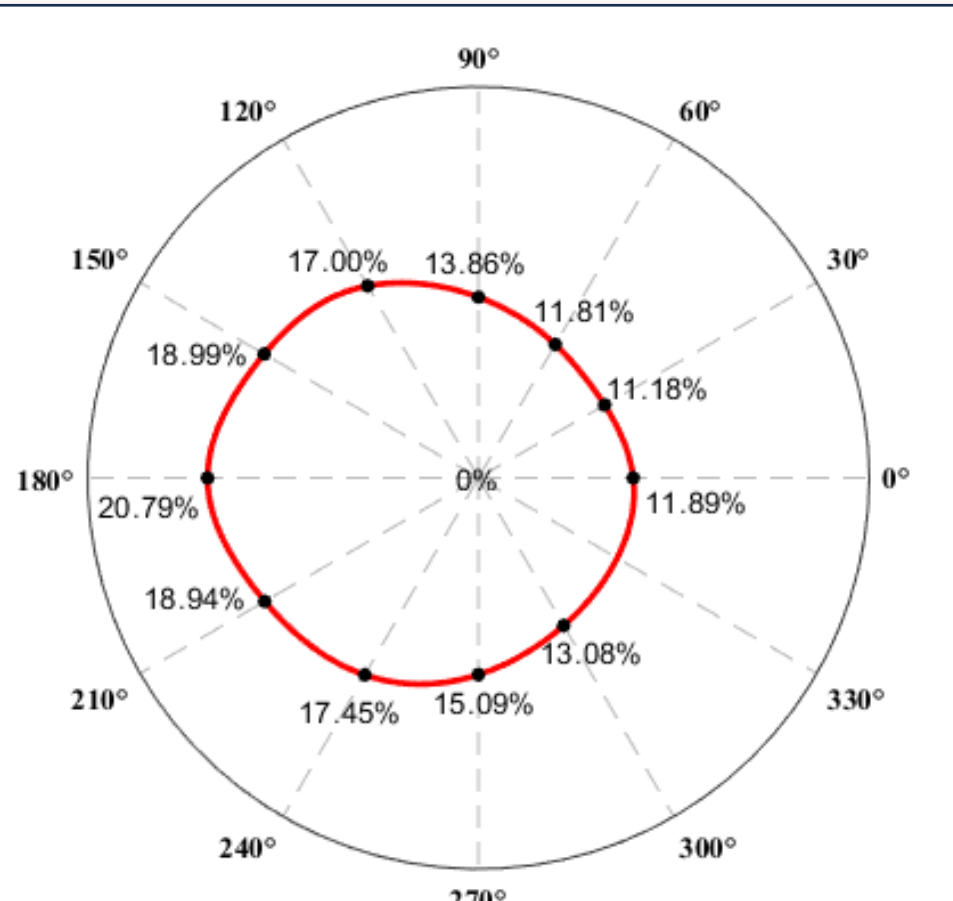


Fig.6: Average Weber fraction